

Chapter 02 Motion in One Dimension

Kinematics

Focus on the **description** of motion, not the **cause** of the motion.

Quantities for Motion

Three concepts:

- Displacement, Distance
- Velocity, Speed
- Acceleration

Position is defined in terms of a **frame of reference**.

Displacement is defined as **the change in position**.

Distance relates to the **actual path traveled**.

Vector and Scalar Quantities

Vector: both magnitude and direction

Scalar: only magnitude

Speed and Velocity

Average speed is total distance divide by total time.

$$v = \frac{d}{t}$$

where v is a scalar quantity.

Average velocity is the **rate** at which the displacement occurs.

$$\vec{v} = \frac{\Delta \vec{x}}{\Delta t}$$

direction will be the same as the direction of the displacement.

Instantaneous Velocity

It is what happens at each **point of time**, the tangent line at a point.

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$

Acceleration

Acceleration is the rate of change of the velocity.

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$$

Instantaneous acceleration is the first time derivative of velocity.

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2}$$

Kinematic Equations

For uniform equations:

$$v = v_0 + at$$

$$\Delta x = \bar{v}t = \frac{1}{2}(v_0 + v)t$$

$$\Delta x = v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a\Delta x$$

Acceleration due to Gravity

Symbolized by g , always directed downward. Free fall is constantly accelerated motion.